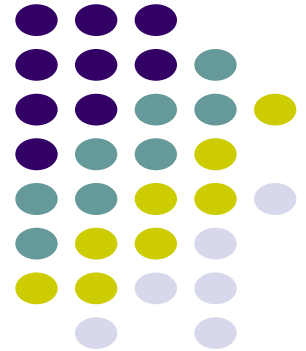


Linux Storage Stack

ECE 469, April 02

Aravind Machiry



Linux Storage Stack

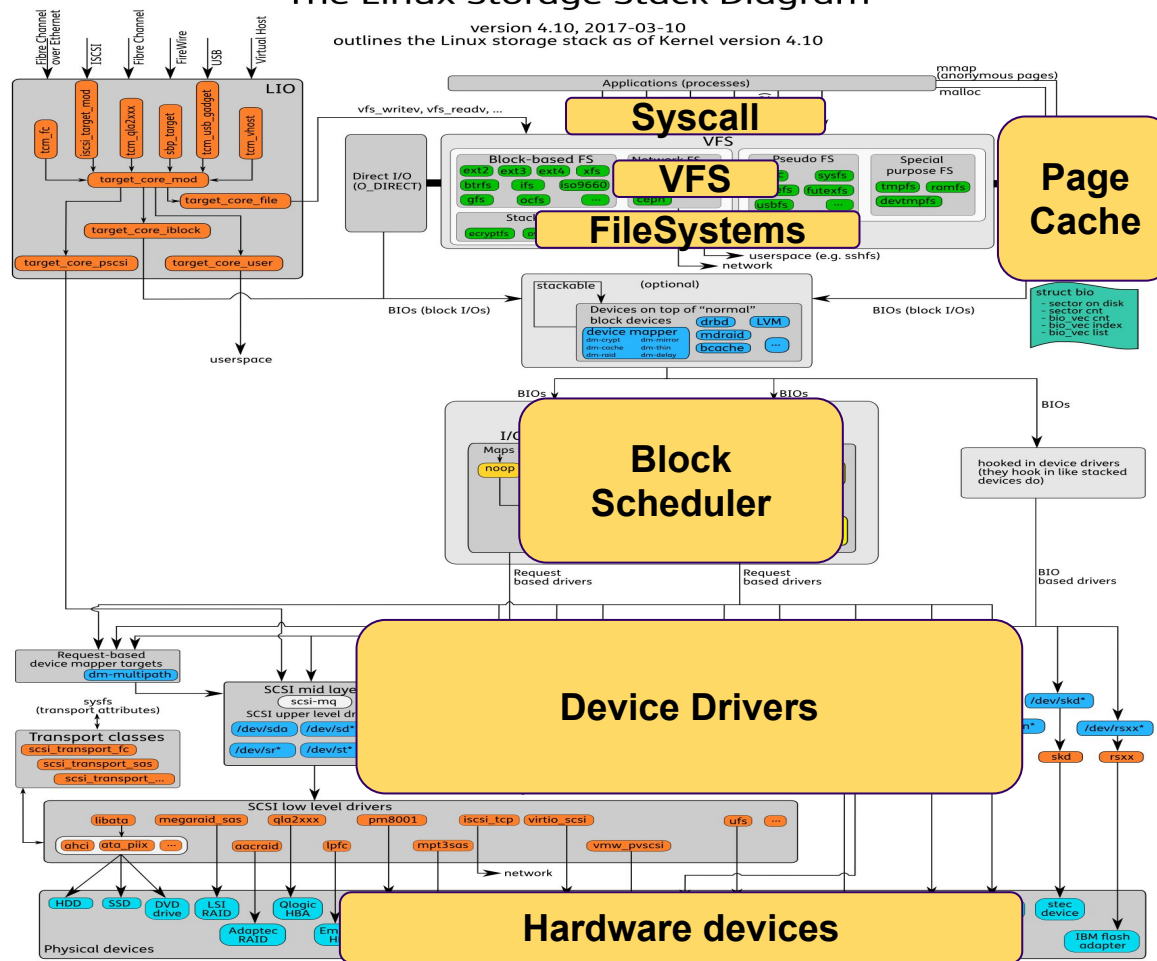
- Exhaustive and Modular



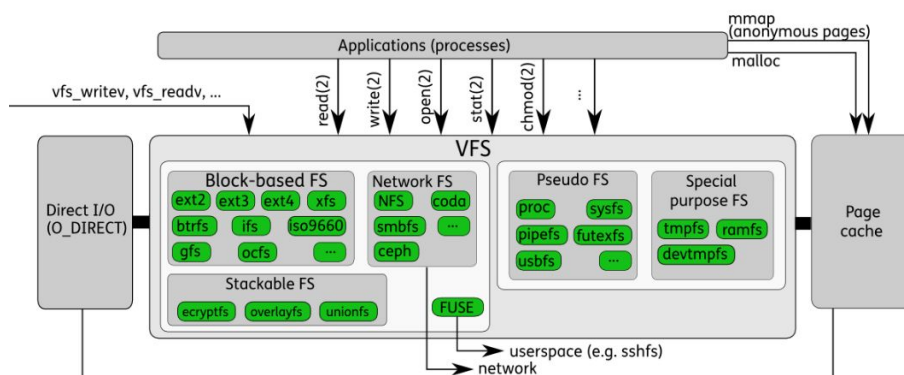
The Linux Storage Stack Diagram
http://www.thomas-krenn.com/en/wiki/Linux_Storage_Stack_Diagram
 Created by Werner Fischer and Georg Schönberger
 License: CC-BY-SA 3.0, see <http://creativecommons.org/licenses/by-sa/3.0>



version 4.10, 2017-03-10
outlines the Linux storage stack as of Kernel version 4.10

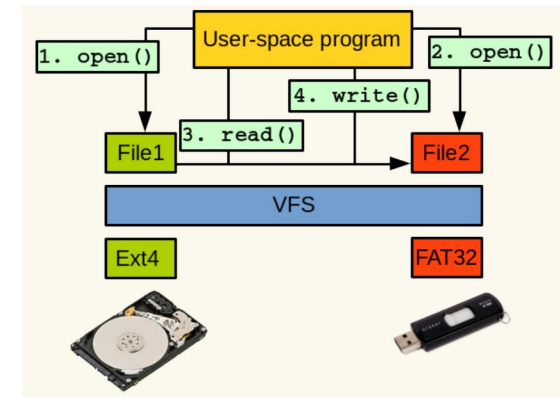


VFS



- Virtual File System (~22K SLOC).
- Everything is a File!!
 - E.g., Network file system! sshfs!?
- ~42 File Systems supported in Linux!!

VFS to Applications

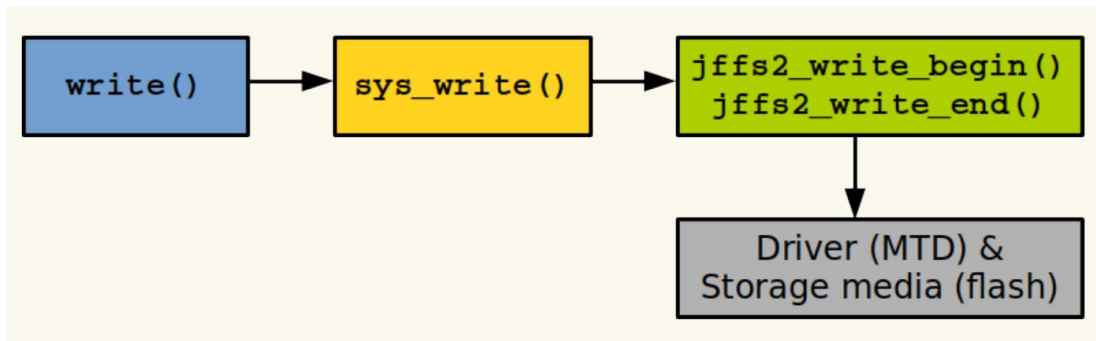


- Common interface for accessing files irrespective of file systems.
- File systems no need to worry about interface to user.

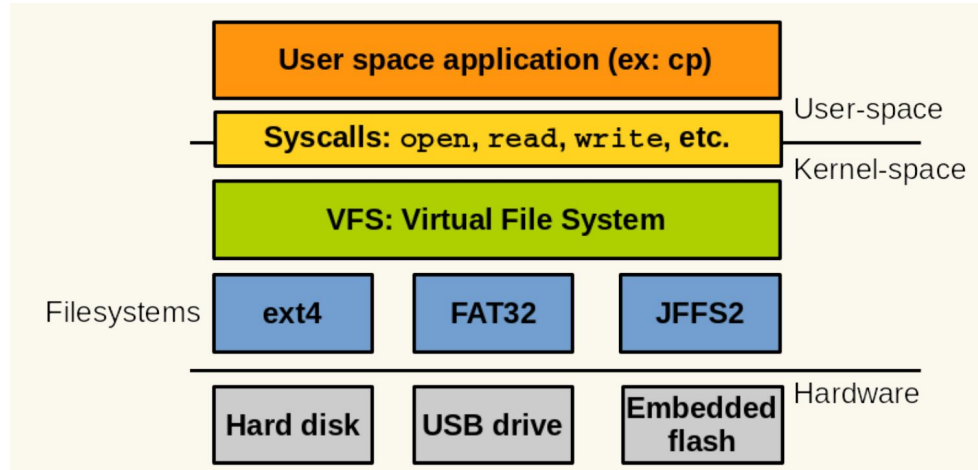
VFS to File System Implementers



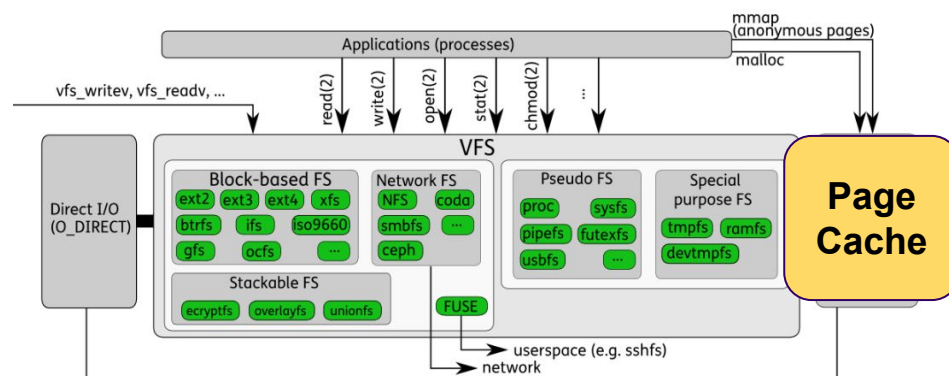
- Exposes common optimization logic. E.g., Page cache, Path lookup.
- Define functions to be implemented by the filesystems.



What does File System Implementers do?



Page Cache



- Reduce Disk IO
- Memory pages maintained by the kernel for storing contents to/from disks.
- Disk block <-> Page



File IO with Page Cache

- *read()*: Serviced by Page Cache!
 - Optimization: Read ahead!
- *write()*: Dirty pages; will be written to disk later!
 - Can loose data!?
- *sync()*: Flush all writes to files.
 - Synchronous

File IO with Page Cache

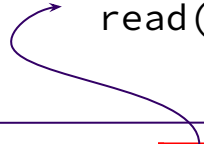


USER

char buf[n]



read()



KERNEL

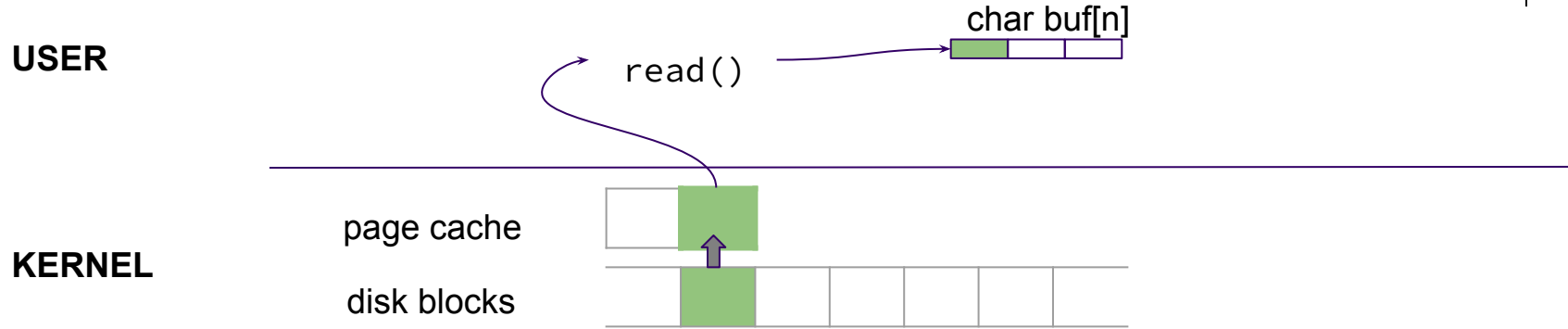
page cache



disk blocks



File IO with Page Cache



Page Cache Implementation



- For each file (inode):
 - Has addr space.
 - File offset -> Page cache.
- For each page:
 - A reference to the file/process.
 - The offset with in the file.

The mmap system call



- Bind virtual memory to file blocks.

```
fd = open("hello.txt", O_RDWR);

// map 4k from offset 0 into virtual address space of the
// process.
char *data = mmap(..,fd, 0);

// read 7th character from file.
char c = data[6];

// write 101th character into file.
data[100] = 'a'
```



Flushing mmap region to file

MSYNC(2)

NAME

`msync` - synchronize a file with a memory map

SYNOPSIS

```
#include <sys/mman.h>
```

```
int msync(void *addr, size_t length, int flags);
```

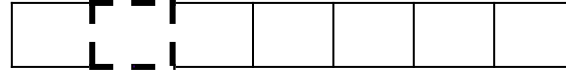
DESCRIPTION

`msync()` flushes changes made to the in-core copy of a file that was mapped in memory. It flushes the part of the file that corresponds to the memory area starting at `addr` and having

Memory RW with Page Cache



USER



mmap

KERNEL

page cache



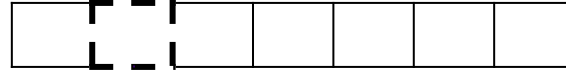
disk blocks



Memory RW with Page Cache



USER



mmap

KERNEL

page cache



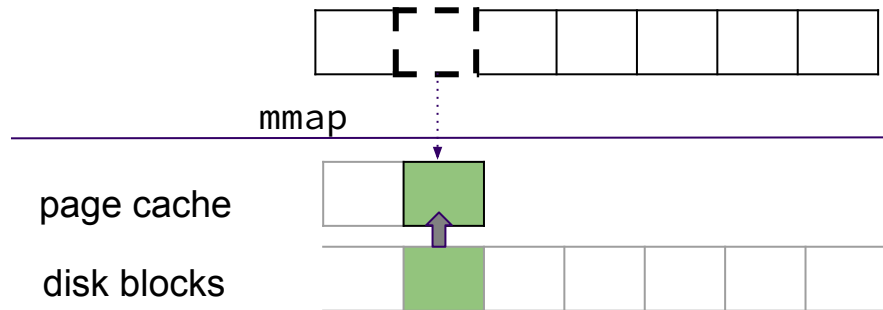
disk blocks



Mmap v/s Explicit IO



- Mmap:
 - No syscalls on each access.
 - Page cache <-> Disk.
 - Dynamic paging.
 - Extra PTEs.
 - Mapping large files? IO Errors?



- File IO
 - Universal.
 - app buffer <-> page cache <-> Disk.

